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Dissolved Gas Analysis System and Method

Abstract

A system for gas analysis is provided. The system includes a gas extraction assembly having an extraction chamber having an upper end, a lower end, and an interior space. The gas extraction assembly further having one or more retainers positioned within the interior space of the extraction chamber. The gas extraction assembly further having gas permeable membrane retained within the interior space of the extraction chamber by the one or more retainers, the gas permeable membrane disposed in S-shaped curves extending from adjacent the upper end to adjacent the lower end within the interior space of the extraction chamber.

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Background/Summary

BACKGROUND

[0001] Electrical power systems including components such as transformers are known to degrade and fail due to aging, thermal and electrical stresses, in which cases the breakdown of insulation material, for example, oil and paper occur. The breakdown of the oil generates certain gases commonly referred to as “fault gases” dissolved in it. Such fault gases may include hydrogen (H.sub.2), carbon monoxide (CO), carbon dioxide (CO.sub.2), methane (CH.sub.4), acetylene (C.sub.2H.sub.2), ethylene (C.sub.2H.sub.4), and ethane (C.sub.2H.sub.6). It is well known in the industry that the amount and nature of these fault gases can be used to identify the type and severity of the fault in the transformer. Therefore, during operation, these dissolved gases may be analyzed or monitored continuously and in real time to evaluate the operational status of the transformer.

SUMMARY

[0002] Described herein is a system and method for online dissolved gas analysis. According to one embodiment, a system for gas analysis is provided. The system includes a gas extraction assembly having an extraction chamber having an upper end, a lower end, and an interior space. The gas extraction assembly further having one or more retainers positioned within the interior space of the extraction chamber. The gas extraction assembly further having gas permeable material retained within the interior space of the extraction chamber by the one or more retainers, the gas permeable material disposed in S-shaped curves extending from adjacent the upper end to adjacent the lower end within the interior space of the extraction chamber.

[0003] According to another embodiment, a gas analysis system is provided that includes a transformer, a gas extraction assembly, a manifold, and an attachment system. The transformer includes an electrical winding and a fluid system comprising insulating fluid. The gas extraction assembly includes an extraction chamber and gas permeable material retained within the extraction chamber. The manifold has a first interface and a second interface, the second interface in fluid communication with the gas extraction assembly, and one or more channels to communicate the fluid between the first and second interfaces. The attachment system includes an adapter pipe, a first interface, and a second interface. The adapter pipe has a first end, a second end, an elongate portion extending from the first end towards the second end, and a bend portion provided adjacent the elongate portion and extending to adjacent the second end. The first interface adjacent the first end of the adapter pipe in fluid communication with the transformer, and the second interface adjacent the second end of the adapter pipe in fluid communication with the first interface of the manifold.

[0004] According to yet another embodiment, a method analyzing dissolved gases in a fluid of a transformer is provided. The method includes providing an adapter pipe having a first end and a second end, the first end in fluid communication with the fluid of the transformer, the second end in fluid communication with a gas analysis system, the adapter pipe having an extension tube provided within the adapter pipe and extending within the adapter pipe between the first end and the second end of the adapter pipe. The method includes supplying the fluid through the extension tube from a first end of the extension tube adjacent the first end of the adapter pipe to a second end of the extension tube adjacent the second end of the adapter pipe, the second end of the extension tube coupled to communicate with a first port of the gas analysis system. The method includes communicating the fluid from the second end of the extension tube, via the first port, into an extraction chamber of the gas analysis system, the extraction chamber having gas permeable material disposed therein. The method includes obtaining the dissolved gas via the gas permeable material, analyzing characteristics of the dissolved gas, and communicating the fluid from the extraction chamber to a second port of the gas analysis system. The method also includes returning the fluid, via the second port, to the second end of the adapter pipe, the second port adjacent the first port such that the supply fluid is supplied adjacent the first end of the adapter pipe and return fluid is returned adjacent the second end of the adapter pipe or via other connection.

[0005] These and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] For a more complete understanding of the present disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

[0007] FIG. 1 is a front perspective view of a gas analysis system according to one embodiment of the present disclosure.

[0008] FIG. 2 is a side view of the gas analysis system according to an embodiment of the present disclosure.

[0009] FIG. 3A is a top perspective view of a retainer of the gas analysis system according to an embodiment of the present disclosure.

[0010] FIG. 3B is partial perspective view of the retainer of the gas analysis system shown in place according to an embodiment of the present disclosure.

[0011] FIG. 3C is cross-sectional view of the retainer of the gas analysis system shown in place according to an embodiment of the present disclosure.

[0012] FIG. 3D is a side perspective view of a supply conduit of the gas analysis system show with the retainers removed.

[0013] FIG. 4 is a layout view of gas permeable material of the gas analysis system according to an embodiment of the present disclosure.

[0014] FIG. 5 is a top view of the gas permeable material of the gas analysis system according to an embodiment of the present disclosure.

[0015] FIG. 6 is a side perspective view of the gas permeable material of the gas analysis system according to an embodiment of the present disclosure.

[0016] FIG. 7 illustrates a support member of the gas analysis system according to an embodiment of the present disclosure.

[0017] FIG. 8 is a back view of the gas analysis system according to an embodiment of the present disclosure.

[0018] FIG. 9 is a perspective view of an attachment system and the gas analysis system according to an embodiment of the present disclosure.

[0019] FIG. 10A is a side cut-away view of the attachment system and the gas analysis system according to another embodiment of the present disclosure.

[0020] FIG. 10B is an exploded side cut-away view of a portion of the attachment system and the gas analysis system according to another embodiment of the present disclosure.

[0021] FIG. 11 is an exploded side cut-away of a portion of the attachment system illustrated in FIG. 10 taken along “C” according to an embodiment of the present disclosure.

[0022] FIG. 12 is a side view of another embodiment of the attachment system and the gas analysis system according to an embodiment of the present disclosure.

[0023] FIG. 13 is a bottom perspective view of an embodiment of the attachment system and the gas analysis system according to the present disclosure.

[0024] FIG. 14 is a top perspective view of the attachment system and the gas analysis system illustrated in FIG. 13 according to the present disclosure.

[0025] FIG. 15 is a side view of a remote attachment system and the gas analysis system according to an embodiment of the present disclosure.

[0026] FIG. 16 is a front perspective view of a mount adapter of the remote attachment system according to an embodiment of the present disclosure.

[0027] FIG. **17** is a back perspective view of the mount adapter of the remote attachment system according to an embodiment of the present disclosure.

[0028] FIG. **18** is a side cut-away view of the mount adapter of the remote attachment system according to an embodiment of the present disclosure.

[0029] FIG. **19** is a flow chart of a method analyzing dissolved gases in a fluid of a transformer according to one embodiment of the present disclosure.

[0030] FIG. **20** is a diagram of an embodiment of a system suitable for implementing various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0031] Described herein is a system and method for dissolved gas analysis. As mentioned above, during operation of transformers, fluid used to cool and insulate the transformer may develop dissolved gas. Analysis of the dissolved gas can provide information that can be used, for example, as an indicator of the operational status of the transformer.

[0032] In order to extract the dissolved gas from the fluid, gas permeable material may be subjected to the transformer fluid to obtain the dissolved gas for analysis. The present disclosure has observed that greater surface exposure of the gas permeable material to the fluid yields increased opportunity for gas to pass through the gas permeable material for analysis. Briefly, the present disclosure provides an extraction chamber, one or more retainers positioned within the extraction the chamber, and gas permeable material retained within the interior space of the extraction chamber by the one or more retainers. The gas permeable material is disposed in S-shaped curves extending from adjacent the upper end to adjacent the lower end within the interior space of the extraction chamber. In addition, transformers are not universal or standardized in all aspects of design and location. Accordingly, the present disclosure provides a manifold that provides for optional configurations for attachment of the present system to access the transformer fluids via attachment systems also disclosed herein. These and other features and advantages are disclosed in greater detail herein.

[0033] FIG. **1** is a front perspective view illustrating of an embodiment of a gas analysis system **100**. The gas analysis system **100** includes a gas extraction assembly **102**, a gas analyzer **103**, and a manifold **104**. The gas analysis system **100** may be attached or in fluid communication with an electrical power system, such as a transformer **106**. The transformer **106**, in this embodiment, may be a core-type, high voltage, three-phase transformer **106**, although other types of transformers or electrical systems may be used. The transformer **106** includes electrical windings **108** disposed in a housing **110**. Fluid **112**, such as dielectric fluid, may be provided in the housing **110** for cooling and insulating the transformer **106** components. As discussed above, dissolved gas **115** may develop in the fluid **112** during operation for various reasons. Accordingly, the transformer **106** includes a valve **114** that is in fluid communication with the fluid **112** in the housing **110** and provides ready access to fluid **112**. In some embodiments, the valve **114** may include any component or fitting to enable access to the fluids **112** within the transformer **106**.

[0034] The gas analyzer **103** includes various gas sensor modules, including but not limited to, the gas sensors based on infrared absorption, thermal conductivity, and electrochemistry, for conducting analysis of the gas collected according to the present system. The gas analyzer **103** may perform any number or manner of gas analysis techniques necessary or desirable to indicate the operational status of the fluid **112** and/or the transformer **106**.

[0035] The gas extraction assembly **102** includes an extraction chamber **116**, which in this embodiment may be a cylindrically shaped cover tube or pipe, although other configurations of the extraction chamber **116** may be used. The extraction chamber **116** extends from an upper end **118** to a lower end **120**. The extraction chamber **116** may be attached to the manifold **104** adjacent the lower end **120** and include a cap **124** or cover coupled adjacent the upper end **118**. The extraction chamber **116** may be attached to the manifold **104** and cap **124** via various attachment systems, such as, but not limited to, by snap rings, threaded attachment, or other attachment systems. The

extraction chamber **116** coupling to the manifold **104** adjacent the lower end **120** and cap **124** adjacent the upper end **118** define an interior space **122** within the extraction chamber **116**.

[0036] A supply conduit **126**, which may be a pipe or tube, is centrally located in the interior space **122** of the extraction chamber **116**. The supply conduit **126** may generally be about a length **128** or height of the extraction chamber **116**. However, the supply conduit **126** has a smaller outer diameter **130**, as defined by an outer wall **131** of the supply conduit **126**, than that of an inner diameter **132** of the extraction chamber **116**, as defined by an inner wall **133** of the extraction chamber **116**. The area between the inner diameter **132** of the extraction chamber **116** and outer diameter **130** of the supply conduit **126** defines a gas extraction area **134** within the interior space **122** of the extraction chamber **116**.

[0037] Referring also to FIG. 2, a side view of the gas analysis system **100** is illustrated. The supply conduit **126** is similarly attached to the manifold **104** adjacent the lower end **120** and the cap **124** adjacent the upper end **118**. In the present embodiment the supply conduit **126** may extend through an upper side **125** of the cap **124**. The extraction chamber **116** may be attached to the manifold **104** and cap **124** via various attachment systems, such as, but not limited to, by snap rings, threaded attachment, or other attachment systems. As can be seen in FIG. 2, the supply conduit **126** includes multiple through holes **136** that allow fluid to be communicated from within a passageway **138** through the supply conduit **126** into the gas extraction area **134**. In the present embodiment, four cross-drilled through holes **136** are provided, however fewer or more openings may be provided and otherwise located in other embodiments.

[0038] Referring also to FIGS. 3A-3D, the gas extraction assembly **102** includes one or more retainers **140A** and **140B**, collectively referred to herein as retainers **140**. In the present embodiment, two retainers **140A** and **140B** are illustrated positioned within the interior space **122** of the extraction chamber **116**, however, in other embodiments, more or fewer retainers **140** may be used. The retainers **140A** and **140B** may be generally flat, round discs or O-rings and may be constructed of polymeric material, such as VITON rubber, while other materials may be used in other embodiments. The retainers **140A** and **140B** have an outer circumference edge **142** and an inner circumference edge **144**. The inner circumference edge **144** defining an opening **146** in the retainers **140A** and **140B**. The retainers **140A** and **140B** include an upper surface **148**, a lower surface **150**, and a plurality of guides **152** or openings that extend through from the upper surface **148** through the lower surface **150**. The plurality of guides **152** are located adjacent the outer circumference edge **142** in this embodiment, but may be located elsewhere in other embodiments. A plurality of slits **154** or cuts are provided in the retainer **140** adjacent each of the guide **152** openings. Each of the slits **154** cut through the retainer **140** and extend linearly from the each of the guides **152** to the outer circumference edge **142**. As will be discussed, the slits **154** provide for ease of assembly, but may be omitted in other embodiments.

[0039] The retainers **140A** and **140B** are positioned in the extraction chamber **116** such that the inner circumference edge **144** of the retainers **140A** and **140B** are positioned adjacent and abuts the outer wall **131** of the supply conduit **126**, and further such that the outer circumference edge **142** of the retainers **140A** and **140B** are positioned adjacent and abuts the inner wall **133** of the extraction chamber **116**. Referring to FIG. 3D, the supply conduit **126** is shown with the retainers removed to illustrate outer grooves **155A** and **155B** configured circumferentially about the outer wall **131** of the supply conduit **126**. The retainers **140A** and **140B** are sized to be positioned in the outer grooves **155A** and **155B**, respectively, of the supply conduit **126** to be retained in position on the supply conduit **126**. In some embodiments, it may be useful to maintain a minimal gap between the retainers **140** and the outer wall **131** of the supply conduit **126** to prevent movement, such as twisting, so the supply conduit **126** does not tug or pull and potentially damage the gas permeable membrane **156**. While two retainers **140A** and **140B** are shown, fewer or more retainers **140** constructed of similar or other materials, may be use and positioned anywhere between the upper end **118** and lower end **120** of the extraction chamber **116**.

[0040] A gas permeable membrane **156** is shown disposed in S-shaped curves extending between adjacent the upper and lower ends **118** and **120** of the extraction chamber **116**. The gas permeable membrane **156**, which may be any gas permeable material, may be a desired length of tubular material that allows dissolved gases in the fluid **112** to permeate into the interior passageway of the tubular gas permeable membrane **156** while preventing the ingress of liquids in the fluid **112**. The gas permeable membrane **156** may be a TEFLON AF (amorphous perfluoropolymer) membrane or other suitable gas permeable material.

[0041] Referring also to FIGS. **1** and **2**, the gas permeable membrane **156** is retained in position within the extraction chamber **116** via retainers **140A** and **140B**. In this embodiment, the gas permeable membrane **156** extends through the guides **152** in the retainers **140A** and **140B** in a series of S-shaped curves. For example, the gas permeable membrane **156** may extend down through one of the guides **152** in the retainer **140A** and continue down through one of the guides **152** in the retainer **140B** that is positioned directly below the guides **152** in the retainer **140A**. Thereafter, the gas permeable membrane **156** curves and returns, extending up, through another, different one of the guides **152** in the retainer **140B** and up through another one of the guides **152** in the retainer **140A** that is positioned directly above the guides **152** in the retainer **140B**, and so on until the gas permeable membrane **156** has been positioned through all the guides **152** in the retainers **140A** and **140B**.

[0042] FIG. **4** depicts the gas permeable membrane **156** laid-out flat to illustrate the S-shaped curves. The gas permeable membrane **156** includes a first end **160** and a second end **162**. FIG. **5** illustrates a top view of the gas permeable membrane **156** positioned in the S-shaped curves **158** and held by the retainers **140A** and **140B**. FIG. **6** is a side perspective view of the gas permeable membrane **156** illustrated in the S-shaped curves **158** with the retainers **140A** and **140B** removed in this view. As can be seen, the S-shaped curves **158** do not intersect one another but instead the gas permeable membrane **156** inner or outer layers merely overlap one another. In some embodiments, the gas permeable membrane **156** may overlap multiple times such as, seven (7) times, while in other embodiments more or fewer overlaps may be used depending on the overall size of the gas extraction assembly **102**, extraction chamber **116**, and desired length of the gas permeable membrane **156** to be used. Where more or fewer loops or layers are desired, the retainers **140** would be appropriately configured.

[0043] It will be appreciated that the gas permeable membrane **156** may be flexible and bend under tension into the S-shaped curves, but when not constrained gas permeable membrane **156** may elastically return to an original or natural shape, such as generally straight. Thus, it may be challenging for a user to assemble the gas permeable membrane **156** in the retainers **140** to achieve the multiple S-shaped curves **158** to form the overall configuration, which may be referred to herein as a “bird-cage” type overall configuration, as shown in FIGS. **1-2** and **5-6**. Accordingly, referring back to FIG. **3**, the slits **154** in the retainers **140** may assist the user in positioning the gas permeable membrane **156** into the guides **152** in the retainers **140**. For example, the user may slide or press a portion of the gas permeable membrane **156** into one of the slits **154** and further into the adjacent guide **152** in order to position that portion of the gas permeable membrane **156** into the guide **152**.

[0044] During assembly, the gas permeable membrane **156** may be threaded or positioned into guides **152** generally across from one another instead of next to one another, in order to prevent crimping the gas permeable membrane **156** and potentially restricting the flow of gases traveling through the gas permeable membrane **156**. To assist the user in consistently threading the gas permeable membrane **156** through the retainers **140**, indicia **164A-164C** may be provided on the upper and/or lower surfaces **148**, **150** of the retainers **140**. For example, the user may begin threading the gas permeable membrane **156** upward, for example, through the guides **152** adjacent the indicia **164A** of the retainers **140A** and **140B**, then loop across and down through the guides **152** adjacent the indicia **164B**, and then loop across and up through the guides **152** adjacent indicia

164C. Thereafter, the user may simply rotate or index to the next or neighboring guide **152** adjacent the guide **152** nearest the previous indicia **164A-C** and repeat the steps of providing the gas permeable membrane **156** in S-shaped curves until the gas permeable membrane **156** has been threaded through all the guides **152** in the retainers **140A** and **140B** to form the overall bird cage configuration. In this embodiment, about 2 meters of gas permeable membrane **156** may be used, however more or less may be used in other embodiments.

[0045] FIG. 7 illustrates a support member **166**, which is also shown in FIGS. 1-3. The support member **166** is a substantially flat O-shaped ring constructed of rubber in this embodiment, but other materials could be used. As can be seen in FIGS. 3B and 3C, the support members **166** may be placed adjacent above and/or below each of the retainers **140A** and **140B**. In this embodiment, the support members **166** include support grooves **167** provided on an inner surface **168** of the support member **166**. The number and location of the support grooves **167** are configured to correspondingly align with the guides **152** in the retainers **140**. The support members **166** are positioned between the gas permeable membrane **156** extending through the guides **152** in the retainers **140** and the inner wall **133** of the extraction chamber **116**. In this position, the support member **166** acts to constrain the elasticity of the gas permeable membrane **156** from expanding into contact with the inner wall **133** of the extraction chamber **116** and retains the gas permeable membrane **156** positioned in the retainers **140**. It will be appreciated that the gas permeable membrane **156** may be fragile and this design may prevent friction or rubbing of the gas permeable membrane **156** with surfaces within the extraction chamber **116**. In addition, the support member **166** further acts to retain the gas permeable membrane **156** positioned in the retainers **140** and prevents the gas permeable membrane **156** from elastically slipping out of or exiting the guides **152** via slits **154** in the retainers **140**. It will be appreciated that general “bird cage” configuration of the present disclosure allows for exposure of more surface area of the gas permeable membrane **156** to the fluid **112** which aids in promoting improved extraction of gas into the gas permeable membrane **156**.

[0046] Returning to FIGS. 3A-3C, the overall size of the of the guides **152** openings may be a matter of design choice. The openings defining the guides **152** are, at a minimum, sized such that the gas permeable membrane **156** may pass through the guide **152** openings. However, to promote improved extraction of gas into the gas permeable membrane **156**, the guide **152** openings may be larger, for example about two (2) to three (3) times or more larger, than the diameter of the gas permeable membrane **156**. As can be seen in FIGS. 3B-3C, this additional space provides for sufficient flow of the fluid **112** between each guide **152** opening and the gas permeable membrane **156** while keeping the flow of fluid **112** sufficiently close to the gas permeable membrane **156** to promote gas extraction. This configuration and design of the gas permeable membrane **156** positioned, via the retainers **140** and support member **166**, in the extraction chamber **116** may further improve, or in some embodiments maximize, the surface area of the gas permeable membrane **156** that is exposed to the fluid **112** as it is flowed through the extraction chamber **116** which promotes improved extraction of the gas in the fluid **112**. FIGS. 3B-3C further illustrates the separation or spreading of the gas permeable membrane **156** when so positioned in the “bird cage” configuration. Thus, while the gas permeable membrane **156** may occasionally overlap and contact along S-shaped curves, the gas permeable membrane **156** is primarily separated from itself or “floats in space” so as to maximize the surface area of the gas permeable membrane **156** exposed to the fluid **112**, which promotes improved gas extraction. For example, in some embodiments, a flow of the fluid **112** through the gas extraction chamber **116** of a gallon per minute, less than a gallon per minute, or perhaps one-tenth (1/10) of a gallon per minute or less, may be sufficient to obtain the desired extraction of gas, while other flow rates may be sufficient in other embodiments.

[0047] Referring again to FIGS. 1 and 2, the gas permeable membrane **156** is coupled at first and second ends **160** and **162** to communicate the diffused gas extracted from the fluid **112** to the gas analyzer **103** via first and second lines **170** and **172**. First and second lines **170** and **172** may be

metal such as stainless-steel piping or tubing. The first and second ends **160**, **162** of the gas permeable membrane **156** and first and second lines **170**, **172** may be connected via fittings **174** attached to a lower side **176** and the upper side of the cap **124**. The fittings **174** may be any fittings or adapters to couple to the first and second ends **160**, **162** of the gas permeable membrane **156** and first and second lines **170**, **172** to communicate the diffused gas therethrough. The cap **124** may include channels or passageways between the fittings **174** to allow for communication of diffused gasses between respective connections on the lower side **176** and upper side **125** of the cap **124**. [0048] The gas analysis system **100** may, in some embodiments, include a pump **178**. The pump **178** is connected to the manifold **104** and configured to promote the flow of the fluid **112** through the gas analysis system **100**. The pump **178** may be any type of pump capable of pumping the fluid **112**, such as diaphragm, centrifugal, or gear pump. As shown in FIG. 2, the manifold **104** includes an inlet channel **180** that extends from a manifold inlet **181** on an angled interface **182** of the manifold **104** to a filter channel **184** in the manifold **104**. The angled interface **182**, which will be described in more detail herein, couples the manifold **104** to receive the fluid **112** from the transformer **106**. The inlet channel **180** and filter channel **184** are fluid tight passageways within the manifold **104**.

[0049] The filter channel **184** extends from a lower side **188** to an upper side **190** of the manifold **104**. A filter **186** is provided in the filter channel **184** near the lower side **188** of the manifold. The filter **186** may be a filter or strainer configured to remove particulates from the fluid **112** before entering the pump **178**. A filter plug **187** is shown connected, such as via threaded attachment, to fluidly seal the filter channel **184** adjacent the lower side **188** of the manifold **104**. The filter plug **187** may be removed to allow access to the filter **186** for replacement or cleaning.

[0050] The pump **178** is attached adjacent the upper side **190** of the manifold **104** at a pump inlet **192**. Fluid **112** drawn from the transformer **106**, via the pump **178**, exits the pump **178** at a pump outlet **194**. The pump outlet **194** is in fluid communication with a transverse channel **196**, via a pump line **198** connected therebetween via couplings **200**, **202**. The pump line **198** may be metal such as stainless-steel piping or tubing. The couplings **200**, **202** may be any fittings or adapters to couple to the pump outlet **194** and the transverse channel **196** to communicate fluid **112** therethrough. Transverse channel **196** is a fluid tight passageway in the manifold **104** that extends from adjacent coupling **202** of pump line **198** to adjacent the supply conduit **126** to the manifold **104**. More specifically, the transverse channel **196** is in fluid communication with the passageway **138** at a lower end of the supply conduit **126**. Thus, the fluid **112** from the transformer **106** is communicated through the manifold **104** and pump **178** into the passageway **138** of the supply conduit **126** and exits through holes **136** of the supply conduit **126** into the extraction chamber **116**. Fluid exiting through holes **136** near upper end **118** of the extraction chamber **116** flows downward toward the lower end **120** of the extraction chamber **116** and reenters the manifold **104** via outlet channel **206**. The outlet channel **206** extends from the lower end **120** of the extraction chamber **116** to a manifold outlet **207** at the angled interface **182** of the manifold **104** where it is returned to the transformer **106**, as well be further discussed below.

[0051] It will be appreciated that the location of the pump **178** and the various fluid communication paths of the manifold **104** may differ in other embodiments and thus, the present embodiment is not limited to the particular configuration described above. For example, in some embodiments, the pump **178** may be omitted, and instead the fluid **112** from the transformer **106** may be communicated through the gas analysis system **100** due to buoyant forces created by temperature gradients in the fluid **112**. In such an embodiment, for example, the filter channel **184**, and the intervening fluid passageway and components may be omitted. These and other embodiments are contemplated and will readily suggest themselves to one skilled in the art in view of the present teachings.

[0052] FIG. 8 is a view of the back side of the gas analysis system **100** of the present disclosure. In this embodiment and the embodiment shown in FIG. 1, the gas analysis system **100** includes a vent

system **210**. The vent system **210** includes a vent line **212** connected and in communication adjacent a first end **214** with the supply conduit **126**. More specifically, the vent line **212** is coupled and in fluid communication with the passageway **138** within the supply conduit **126**. The location of the connection of the vent line **212** to the supply conduit **126**, may in this embodiment, represent the highest point of the fluid **112** flow path through the gas analysis system **100**. In this embodiment, the vent line **212** is connected at a second end **216** to the upper side **190** of the manifold **104**. The vent line **212** communicates at the second end **216** with a vent channel **218** that extends within the manifold **104** from the upper side **190** to the lower side **188** of the manifold **104**. A sampling port **220** may be connected, such as via a threaded connection, to the lower side **188** of the manifold **104** and be in fluid communication with the vent channel **218**. The sampling port **220** may be variously configured, such as via elbows, unions, caps, and ports, such as a sampling and/or air purge port **222**. Sampling and/or air purge port **222** is openable to the surrounding area.

[0053] It will be appreciated that when the gas analysis system **100** is initially connected to the transformer **106**, serviced, or otherwise, the manifold **104** and gas extraction assembly **102** may contain air that may need to be purged out of the gas analysis system **100** to prevent air from entering the transformer **106** or for other reasons. It can be seen that, by opening the sampling and/or air purge port **222**, the vent system **210** allows the static head pressure of the fluid **112** to flow into the gas analysis system **100** which causes air to be expelled from the gas analysis system **100**. The pump **178** may be optionally operated at low speed to assist during this process. Further, periodically transformers **106** may require that samples of the fluid **112** or oil in the transformer **106** be taken for analysis. The fluid **112** samples may be retrieved via the sampling and/or air purge port **222**.

[0054] While some transformers **106** may include ports, such as valve **114**, specifically provided on the transformer **106** for connecting systems, such as the gas analysis system **100**, other transformers **106** may only include drain or other valves or connection points. FIGS. **9** and **10A-10B** illustrate an attachment system **300** for connecting the gas analysis system **100** to a drain valve **302** of the transformer **106**. Although the drain valve **302** is illustrated as a gate or ball valve, other well-known ports or valves may be used in other embodiments. The drain valve **302** is connected to and in fluid communication with the fluid **112** within the transformer **106**.

[0055] In the present embodiment, the gas extraction assembly **102** is oriented or extends generally vertically relative to manifold **104** and a horizontal plane **318**. Such orientation of the gas analysis system **100** may be referred to as a “vertical” orientation. The present disclosure is not limited to this orientation and other positioning may be used in other embodiments. In the “vertical” orientation shown in the present embodiment, it can be seen that the angled interface **182** of the manifold **104** is angled relative to the horizontal plane **318** at an angle, such as 45 degrees (45°). Thus, the angled interface **182** may be said to be at an incline or acute angle relative to the horizontal plane **318**.

[0056] The attachment system **300** includes an adapter fitting **304** connected at a first end **306** adjacent the drain valve **302** and connected at a second end **308** to the angled interface **182** of the manifold **104** of the gas analysis system **100**. The adapter fitting **304** may be a tubular pipe or fitting having an outer surface **310** defining a fluid passageway **312** extending between the first and second ends **306** and **308** of the adapter fitting **304**. An interface **314** is provided at the second end **308** of the adapter fitting **304** to mate with the angled interface **182** of the manifold **104**. The adapter fitting **304** extends as an elongate section from the first end **306** toward a bend portion **316** adjacent the second end **308** at a bend of approximately 45 degrees (45°). The bend portion **316** allows the adapter fitting **304** to extend from the first end **306** horizontally, relative to the horizontal plane **318**, straight from the drain valve **302** and angles at the bend portion **316** so that the interface **314** at the second end **308** angularly mates with the angled interface **182** of the manifold **104**. The interface **314** of the adapter fitting **304** is attachable to angled interface **182** of the manifold **104** via screws, bolts, threaded or other well-known systems for accomplishing a

mechanical connection and fluid tight connectors such as O-rings for a liquid tight connection. [0057] As can be seen in FIGS. **10A-10B**, the interface **314** includes an interface inlet **320** and an interface outlet **322**. When the interface **314** is connected to the angled interface **182** of the manifold **104**, the interface inlet **320** is in fluid communication with the manifold outlet **207** and the interface outlet **322** is in fluid communication with the manifold inlet **181**. The interface outlet **322** may include a hose nipple **324** captured between the interface **314** of the adapter fitting **304** and angled interface **182** of the manifold **104** and extending from the interface outlet **322** into the fluid passageway **312**. The interface **314** of the adapter fitting **304** may include various rubber gaskets or O-rings **325A** and **325B** on interface inlet and outlet **320** and **322** to promote a fluid tight seal. An extension tube **326** or pipe may be connected to the hose nipple **324** and extend through the fluid passageway **312** to adjacent the first end **306** of the adapter fitting **304**. The extension tube **326** allows fluid to flow within the extension tube **326** from the hose nipple **324** to adjacent the first end **306** of the adapter fitting **304**. As can be seen, the extension tube **326** has a smaller outer diameter **328** than the larger inner diameter **330** of the adapter fitting **304**. In this embodiment, the extension tube **326** may be a flexible rubber (e.g. Viton®) or plastic (e.g. polytetrafluoroethylene) tube or hose, but may be rigid, for example metal, such as stainless steel, brass, carbon steel, in other embodiments.

[0058] FIG. **11** is an expanded cut-away portion shown in FIG. **10A** taken along “C” illustrating one embodiment of the coupling of the attachment system **300** to the drain valve **302**. In this embodiment, a union receiver **340** is provided that includes a first threaded portion **342** and a second threaded portion **344**. The first threaded portion **342** is connected to drain valve **302** adjacent a first end **346** of the union receiver **340**. The first end **306** of adapter fitting **304** is positioned within the union receiver **340**. The second threaded portion **344** of the union receiver **340** is connected to a threaded portion **348** of a union nut **350** that is provided to surround the adapter fitting **304** adjacent the first end **306**. A retaining ring **352** is located adjacent the threaded interface of the union receiver **340** and union nut **350** to transfer thrust from the union nut **350** to the adapter fitting **304** driving the adapter fitting **304** into engagement with the union receiver **340**. [0059] As can be seen, the union receiver **340** includes a shoulder **358** that provides a primary fluid sealing point of the union receiver **340** with the first end **306** of the adapter fitting **304**. The shoulder **358** also provides a ramp or slope that allows the extension tube **326** to extend beyond the interface of the union receiver **340** and first end **306** of the adapter fitting **304** without getting caught or crimped at that point. In this embodiment, the adapter fitting **304** may include a notch **354** adjacent the first end **306** of the adapter fitting **304** for receiving an O-ring **356**, such as a rubber gasket O-ring, to provide for further fluid sealing of the connection.

[0060] As can be seen, the attachment system **300** is configured such that the first end **306** generally terminates vertically, relative to the horizontal plane **318**, while the second end **308**, at the attachment interface **314**, generally terminates at an angle, such as 45 degrees (45°) relative to the horizontal plane **318**, to angularly mate with the angled interface **182** of the manifold **104**.

[0061] In practice, when the drain valve **302** is opened, fluid **112** from the transformer **106** is drawn into the extension tube **326** adjacent the first end **306** of the adapter fitting **304**. The fluid **112** traverses the extension tube **326**, via static pressure, the pump **178**, or otherwise, exits the attachment system **300**, via the hose nipple **324** and the interface outlet **322**, and enters the gas analysis system **100**, via manifold inlet **181**. After analysis by the gas analysis system **100**, the fluid **112** exits the gas analysis system **100**, via manifold outlet **207**, and enters the attachment system **300**, via interface inlet **320**, and flows into the fluid passageway **312** within the adapter fitting **304**.

[0062] Thus, according to the present disclosure, the fluid **112** is drawn into the extension tube **326** which is located closer to the transformer **106** than the interface **314** adjacent the second end **308** of the adapter fitting **304**. It will be appreciated that there may be advantages, in some embodiments, for having the gas analysis system **100** and the fluid **112** being sampled, in close physical proximity to the transformer **106** so that the fluid **112** from the transformer **106** being analyzed is most

representative of the fluid **112** in the transformer **106**. Further, once the fluid **112** has been tested by the gas analysis system **100**, it is returned to the fluid passageway **312** adjacent the second end **308** of the adapter fitting **304**. Since the gas analysis system **100** removes certain amounts of the gas in the fluid **112** for testing, the returned fluid **112** is no longer representative of the gas content of the fluid **112** as it exists in the transformer **106**. By drawing the fluid **112** to be sampled or tested at a point closer to the transformer **106**, a more representative sample of fluid and consequently a more accurate test may be conducted in contrast to drawing the fluid **112** to be sampled from the same location as the returned, gas extracted fluid **112**. Furthermore, as the returned or tested fluid **112** enters the fluid passageway **312**, the fluid **112** remixes with the fluid **112** from the transformer **106** forced into the fluid passageway **312**. Thus, even after time, the fluid **112** drawn into the extension tube **326** continues to be representative of the fluid **112** in the transformer **106**, which promotes more accurate ongoing testing of the fluid **112**.

[0063] While the gas analysis system **100** is generally shown in FIGS. **9** and **10** positioned horizontally, relative to the horizontal plane **318**, extending from the drain valve **302** of the transformer **106**, in some instances it may be useful to provide the gas analysis system **100** at other locations due to the space available in surrounding areas or for other reasons. FIG. **12** illustrates another implementation of the present disclosure. In this embodiment, a first pipe **380** is shown attached at a first end **382** to the drain valve **302**. The first pipe **380** extends from the drain valve **302** horizontally, relative to the horizontal plane **318**, and connects to an elbow **384** at a second end **386** of the first pipe **380**. A second pipe **388** is attached at a first end **390** of the second pipe **388** to the elbow **384** and at a second end **392** of the second pipe **388** to the attachment system **300**. The connection between attachment system **300** and the second end **392** of the second pipe **388** is accomplished similar to that described in FIGS. **9-11** with regard to the connection of the attachment system **300** and the drain valve **302**.

[0064] As can be seen in FIG. **12**, the attachment system **300** is oriented generally vertically, relative to the horizontal plane **318**, instead of generally horizontally, as shown in FIGS. **9** and **10**. Due to the angular configuration of the angled interface **182** of the manifold **104** and angled interface **314**, the adapter fitting **304** of the attachment system **300** may simply be re-oriented to reposition the gas analysis system **100** in such position or location while still maintaining the “vertical” orientation of the gas analysis system **100**.

[0065] In this embodiment, the extension tube **326** may extend from adjacent the manifold **104** through the attachment system **300**, second pipe **388**, elbow **384**, first pipe **380**, and terminate adjacent the drain valve **302**. In this manner, the fluid **112** is drawn into the extension tube **326** which is located closer to the transformer **106** than the interface **314** adjacent the second end **308** of the adapter fitting **304**. As mentioned above, this provides for the fluid **112** being sampled to be in close physical proximity to the transformer **106** so that the fluid **112** from the transformer **106** being analyzed is most representative of the fluid **112** in the transformer **106**. Thus, even though the gas analysis system **100** is located farther away or remote from the transformer **106**, the present system continues to provide for sampling of the fluid **112** nearer the transformer **106**, which may provide advantages as discussed above. It will be appreciated that the present disclosure is not so limited and, in other embodiments, the extension tube **326** may extend more or less far, such as only into the second pipe **388** for example.

[0066] Referring to FIGS. **13** and **14** another embodiment of the present disclosure is shown. It will be appreciated that the weight of the gas analysis system **100** and attachment system **300** may place stress on the drain valve **302**, such as small size valves, or other attachment points of the transformer **106**. As shown in FIGS. **13** and **14**, the present disclosure provides a support system **400** to support the weight of the gas analysis system **100** and attachment system **300**. The support system **400** may include a support arm **402**, which may be a pipe or other piece, attached at an upper end **404** to the lower side **188** of the manifold **104** and at a lower end **406** to a floor fitting **408** of the support system **400**.

[0067] In this embodiment, the manifold **104** includes a socket **410** which may be integrally formed or cast into the manifold **104** and configured to receive the upper end **404** of the support arm **402**. The support arm **402** may be secured to the socket **410** of the manifold **104** at the upper end **404** and the floor fitting **408** at the lower end **406** via tensioning screws or otherwise. The floor fitting **408**, such as a floor flange or rail fitting, is configured for attachment or engagement with a support surface, such as the ground or concrete pad, adjacent the transformer **106**. In this manner, the weight or load of the gas analysis system **100** and attachment system **300** is transferred to the support surface relieving pressure or mechanical strain on the drain valve **302** and the attachment system **300** connection or interface **314**.

[0068] While the embodiments shown in FIGS. **13** and **14** show the attachment system **300** and gas analysis system **100** in the generally horizontal position, the support system **400** may be readily adapted, such as by providing a longer support arm **402** for supporting the gas analysis system **100** when provided in the vertical position as shown in FIG. **12**. Depending on the desired application, it will be appreciated that additional pipe or coupling pieces or other makeup fittings may be used in various configurations in other embodiments as will readily suggest themselves to one skilled in the art, all of which are within the spirit and scope of the present disclosure.

[0069] The present disclosure further contemplates deployment of the gas analysis system **100** in locations more remote than is practicable using only the attachment system **300** and pipes and fittings shown in FIGS. **9**, **10**, and **12**. Instead, as shown in FIG. **15**, a remote attachment system **430** is provided. The remote attachment system **430** includes a mount adapter **432**, a supply line **434**, and a return line **436**. The mount adapter **432** is shown attached to the angled interface **182** of the manifold **104** of the gas analysis system **100**. The mount adapter **432** is coupled to the angled interface **182** via screws **438**, but may be otherwise attached in other embodiments.

[0070] FIGS. **16-18** illustrate various views of the mount adapter **432**. The mount adapter **432** includes a return port **440** that is in fluid communication with the manifold outlet **207** when the mount adapter **432** is coupled to the angled interface **182**. A return channel **442** extends within the mount adapter **432** between the return port **440** and a return outlet **444**. The return outlet **444** is provided adjacent an adapter rear side **446** substantially opposite the return port **440**. The return line **436** fluidly couples to the return outlet **444**.

[0071] The mount adapter **432** also includes a supply port **448** that is in fluid communication with the manifold inlet **181** when the mount adapter **432** is coupled to the angled interface **182**. A supply channel **450** extends within the mount adapter **432** between the supply port **448** and a supply outlet **452**. The supply outlet **452** is provided adjacent the adapter rear side **446** substantially opposite the supply port **448**. The supply line **434** fluidly couples to the supply outlet **452**.

[0072] First ends **454**, **456** of the return and supply lines **436**, **434**, respectively, may fluidly couple to a remote interface **458**. The remote interface **458** may be configured to couple directly to the drain valve **302** or other connection points of the transformer **106** or via the attachment system **300**. For example, the first ends **454**, **456** of the return and supply lines **436**, **434**, respectively, may fluidly couple to a single drain valve **302** or separate oil inlet and oil outlets of the transformer **110**.

[0073] The remote attachment system **430**, like the various other systems and components described herein, such as the attachment system **300** and gas analysis system **100**, may be constructed of brass, steel, stainless steel, polymeric, rubber, or other well-known materials as desired or as described herein.

[0074] Referring to FIG. **19**, a method **500** for analyzing dissolved gases in a fluid of a transformer is shown. The method includes, at box **520**, providing an adapter pipe having a first end and a second end, the first end in fluid communication with the fluid of the transformer, the second end in fluid communication with a gas analysis system, the adapter pipe having an extension tube provided within the adapter pipe and extending within the adapter pipe between the first end and the second end of the adapter pipe. The method includes, at box **522**, supplying the fluid through the extension tube from a first end of the extension tube adjacent the first end of the adapter pipe to

a second end of the extension tube adjacent the second end of the adapter pipe, the second end of the extension tube coupled to communicate with a first port of the gas analysis system. The method includes, at box **524**, communicating the fluid from the second end of the extension tube, via the first port, into an extraction chamber of the gas analysis system, the extraction chamber having gas permeable material disposed therein. The method includes, at box **526**, obtaining the dissolved gas via the gas permeable material. The method includes, at box **528**, analyzing characteristics of the dissolved gas. The method includes, at box **530**, communicating the fluid from the extraction chamber to a second port of the gas analysis system. The method includes, at box **532**, returning the fluid, via the second port, to the second end of the adapter pipe, the second port adjacent the first port such that the supply fluid is supplied adjacent the first end of the adapter pipe and return fluid is returned adjacent the second end of the adapter pipe. In some embodiments, the fluid may be returned to fitting not adjacent the first end, such as higher or lower on the transformer or elsewhere for communication to the transformer.

[0075] FIG. **20** is a diagram of an embodiment of a system **900** that includes a processor **910** suitable for implementing one or more embodiments disclosed herein, e.g., the gas analysis system **100** and/or gas analyzer **103**. The processor **910** may control the overall operation of the system **900**. In some embodiments, processor **910** may be a microcontroller including components described below, for example, RAM **930**, ROM **940**, and/or I/O **960**. Although only one processor is shown, multiple processors may be employed in practice.

[0076] In addition to the processor **910** (which may be referred to as a central processor unit or CPU), of the system **900** might include network connectivity devices **920**, random access memory (RAM) **930**, read only memory (ROM) **940**, secondary storage **950**, and input/output (I/O) devices **960**. These components might communicate with one another via a bus **970**. In some cases, some of these components may not be present or may be combined in various combinations with one another or with other components not shown. These components might be located in a single physical entity or in more than one physical entity. Any actions described herein as being taken by the processor **910** might be taken by the processor **910** alone or by the processor **910** in conjunction with one or more components shown or not shown in the drawing, such as a digital signal processor (DSP) **980**. Although the DSP **980** is shown as a separate component, the DSP **980** might be incorporated into the processor **910**.

[0077] The processor **910** executes instructions, codes, computer programs, or scripts that it might access from the network connectivity devices **920**, RAM **930**, ROM **940**, or secondary storage **950** (which might include various disk-based systems such as hard disk,, or optical disk). While only one CPU **910** is shown, multiple processors may be present. Thus, while instructions may be discussed as being executed by a processor, the instructions may be executed simultaneously, serially, or otherwise by one or multiple processors. The processor **910** may be implemented as one or more CPU chips and may be a hardware device capable of executing computer instructions.

[0078] The network connectivity devices **920** may take the form of modems, modem banks, Ethernet devices, universal serial bus (USB) interface devices, serial interfaces, token ring devices, fiber distributed data interface (FDDI) devices, wireless local area network (WLAN) devices, radio transceiver devices such as code division multiple access (CDMA) devices, global system for mobile communications (GSM) radio transceiver devices, universal mobile telecommunications system (UMTS) radio transceiver devices, long term evolution (LTE) radio transceiver devices, worldwide interoperability for microwave access (WiMAX) devices, controller area network (CAN), domestic digital bus (D2B), and/or other well-known devices for connecting to networks. These network connectivity devices **920** may enable the processor **910** to communicate with the Internet or one or more telecommunications networks or other networks from which the processor **910** might receive information or to which the processor **910** might output information. The network connectivity devices **920** might also include one or more transceiver components **925** capable of transmitting and/or receiving data wirelessly.

[0079] The RAM **930** might be used to store volatile data and perhaps to store instructions that are executed by the processor **910**. The ROM **940** is a non-volatile memory device that typically has a smaller memory capacity than the memory capacity of the secondary storage **950**. ROM **940** might be used to store instructions and perhaps data that are read during execution of the instructions. Access to both RAM **930** and ROM **940** is typically faster than to secondary storage **950**. The secondary storage **950** is typically comprised of one or more disk drives or tape drives and might be used for non-volatile storage of data or as an over-flow data storage device if RAM **930** is not large enough to hold all working data. Secondary storage **950** may be used to store programs that are loaded into RAM **930** when such programs are selected for execution.

[0080] The I/O devices **960** may include liquid crystal displays (LCDs), touch screen displays, keyboards, keypads, switches, dials, mice, track balls, voice recognizers, card readers, paper tape readers, printers, video monitors, electrical connectors, electrical excitation sources, electrical measurement devices, switching devices, relay devices, or other well-known input/output devices. Also, the transceiver component **925** might be considered to be a component of the I/O devices **960** instead of or in addition to being a component of the network connectivity devices **920**.

[0081] While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods may be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted or not implemented.

[0082] Also, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component, whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

Claims

1. A gas analysis system, comprising: a gas extraction assembly comprising: an extraction chamber having an upper end, a lower end, and an interior space; one or more retainers positioned within the interior space of the extraction chamber; and gas permeable membrane retained within the interior space of the extraction chamber by the one or more retainers, the gas permeable membrane disposed in S-shaped curves extending from adjacent the upper end to adjacent the lower end within the interior space of the extraction chamber.
2. The gas analysis system of claim 1, further comprising: a transformer comprising an electrical winding and a fluid system comprising insulating fluid; and a manifold having first interface in fluid communication with the transformer, a second interface in fluid communication with the extraction chamber, and one or more channels to communicate the fluid between the first and second interfaces.
3. The gas analysis system of claim 2, wherein the first and second interfaces of the manifold each include a supply channel and a return channel.
4. The gas analysis system of claim 3, wherein the extraction chamber further comprises: an outer cover defining the interior space of the extraction chamber; and a supply conduit centrally located in the extraction chamber, the supply conduit in fluid communication with the supply channel of the second interface of the manifold at a first end adjacent the lower end of the chamber and an output port located adjacent the upper end of the extraction chamber, wherein the output port is in

fluid communication with the interior space of the extraction chamber.

5. The gas analysis system of claim 4, wherein the each of the one or more retainers is further defined as an O-shaped ring having an outer circumference edge, an inner circumference edge defining an inner opening, an upper surface, a lower surface, and a plurality of guide openings extending from the upper surface through to the lower surface and located adjacent and between the outer and inner circumference edges, wherein the outer circumference edge is adjacent and frictionally engages an interior wall of the chamber and the inner circumference edge is adjacent and frictionally engages an outer wall of the supply conduit, and wherein the gas permeable material passes through the plurality of guide openings.

6. The gas analysis system of claim 5, wherein the gas permeable membrane is further defined as gas permeable tubing having a first end and a second end.

7. The gas analysis system of claim 6, further comprising: a gas analyzer having one or more sensors for analyzing characteristics of gas extracted from the fluid, the gas analyzer in fluid communication with the first and second ends of the gas permeable tubing to receive the gas extracted from the fluid for analysis by the gas analyzer.

8. The gas analysis system of claim 2, further comprising one or more pumps in fluid communication with the manifold to promote the communication of fluid between the first and second interfaces.

9. The gas analysis system of claim 2, wherein the second interface of the manifold is located on an upper surface of the manifold, wherein the first interface of the manifold is located on a first side of the manifold, and wherein the first side of the manifold is angled at approximately 45 degrees relative to a horizontal plane of the upper surface.

10. A gas analysis system, comprising a transformer having an electrical winding and a fluid system comprising insulating fluid; a gas extraction assembly having an extraction chamber and gas permeable membrane retained within the extraction chamber; a manifold having a first interface and a second interface, the second interface in fluid communication with the gas extraction assembly, and one or more channels to communicate the insulating fluid between the first and second interfaces; and an attachment system comprising: an adapter pipe having a first end, a second end, an elongate portion extending from the first end towards the second end, and a bend portion provided adjacent the elongate portion and extending to adjacent the second end, a first interface adjacent the first end of the adapter pipe in fluid communication with the transformer, and a second interface adjacent the second end of the adapter pipe in fluid communication with the first interface of the manifold.

11. The gas analysis system of claim 10, wherein the second interface of the attachment system includes an inlet port and an outlet port, the inlet port in fluid communication with an outlet port of the second interface of the manifold, the outlet port in fluid communication with an inlet port of the second interface of the manifold.

12. The gas analysis system of claim 11, wherein the attachment system further comprises an extension tube provided within the adapter pipe and attached to the outlet port of the second interface of the attachment system at a first end of the extension tube, and wherein the extension tube extends to a second end of the extension tube adjacent the first end of the adapter pipe.

13. The gas analysis system of claim 12, wherein the extension tube has an outer diameter that is less than an inner diameter of the adapter pipe, and wherein the extension tube is flexible plastic, rubber or metallic, or a combination thereof.

14. The gas analysis system of claim 10, where a horizontal plane extends between first and second sides of the manifold, and wherein the gas extraction assembly is mounted on an upper side of the manifold such that the extraction chamber extends vertically relative to the horizontal plane.

15. The gas analysis system of claim 14, wherein the first interface of the manifold is located on the first side of the manifold, and wherein the first interface is angled at approximately 45 degrees relative to the horizontal plane.

16. The gas analysis system of claim 15, wherein in a first position of the attachment system the elongate portion of the adapter pipe is oriented substantially parallel to the horizontal plane, and wherein in a second position of the attachment system the elongate portion of the adapter pipe is oriented substantially perpendicular to the horizontal plane, and wherein the bend portion of the adapter pipe bends such that the second interface adjacent the second end of the adapter pipe mates with the angled first interface of the manifold when the attachment system is in either the first or second positions.

17. The gas analysis system of claim 16, wherein when the attachment system is in the first or second positions, the manifold is positioned such that the horizontal plane is substantially parallel to an adjacent ground surface and the gas extraction assembly extends vertically relative to the horizontal plane and the ground surface.

18. The gas analysis system of claim 10, wherein the attachment system fluidly communicates with a fluid in the transformer via one or more of: attachment of the first interface adjacent the first end of the adapter pipe to a drain valve of the transformer; or attachment to a fluid access point of the transformer via one or more pipe or coupling pieces, pipe fittings, elbows in communication with the first interface adjacent the first end of the adapter pipe.

19. A method analyzing dissolved gases in a fluid of a transformer, the method comprising: providing an adapter pipe having a first end and a second end, the first end in fluid communication with the fluid of the transformer, the second end in fluid communication with a gas analysis system, the adapter pipe having an extension tube provided within the adapter pipe and extending within the adapter pipe between the first end and the second end of the adapter pipe; supplying the fluid through the extension tube from a first end of the extension tube adjacent the first end of the adapter pipe to a second end of the extension tube adjacent the second end of the adapter pipe, the second end of the extension tube coupled to communicate with a first port of the gas analysis system; communicating the fluid from the second end of the extension tube, via the first port, into an extraction chamber of the gas analysis system, the extraction chamber having gas permeable membrane disposed therein; obtaining the dissolved gas via the gas permeable membrane; analyzing characteristics of the dissolved gas; communicating the fluid from the extraction chamber to a second port of the gas analysis system; and returning the fluid, via the second port, to the second end of the adapter pipe, the second port adjacent the first port such that the supply fluid is supplied adjacent the first end of the adapter pipe and return fluid is returned adjacent the second end of the adapter pipe.
